

Ex. No. : 02

Date:

Register No.: 231701017

Name: HARISH BABU R

A PYTHON PROGRAM TO IMPLEMENT SIMPLE LINEAR REGRESSION USING LEAST SQUARE METHOD

Aim:

To implement **Simple Linear Regression** using the **Least Square Method** in Python and analyze the relationship between one independent variable and one dependent variable from the Iris dataset.

Requirements:

Jupyter Notebook with Python, Iris.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from google.colab import files

uploaded = files.upload()

df = pd.read_csv('Iris.csv')

X = df['SepalLengthCm'].values

Y = df['PetalLengthCm'].values
```

```
n = len(X)
```

```
sum_x = np.sum(X)
```

```
sum_y = np.sum(Y)
```

```
sum_xy = np.sum(X * Y)
```

```
sum_x2 = np.sum(X**2)
```

```
b = (n * sum_xy - sum_x * sum_y) / (n
```

```
* sum_x2 - sum_x**2)
```

```
a = (sum_y - b * sum_x) / n
```

```
print("Slope (b):", b)
```

```
print("Intercept (a):", a)
```

```
Y_pred = a + b * X
```

```
sorted_indices = np.argsort(X)
```

```
X_sorted = X[sorted_indices]
```

```
Y_pred_sorted =
```

```
Y_pred[sorted_indices]
```

```
plt.figure()
```

```
plt.scatter(X, Y)
```

```
plt.plot(X_sorted, Y_pred_sorted)
```

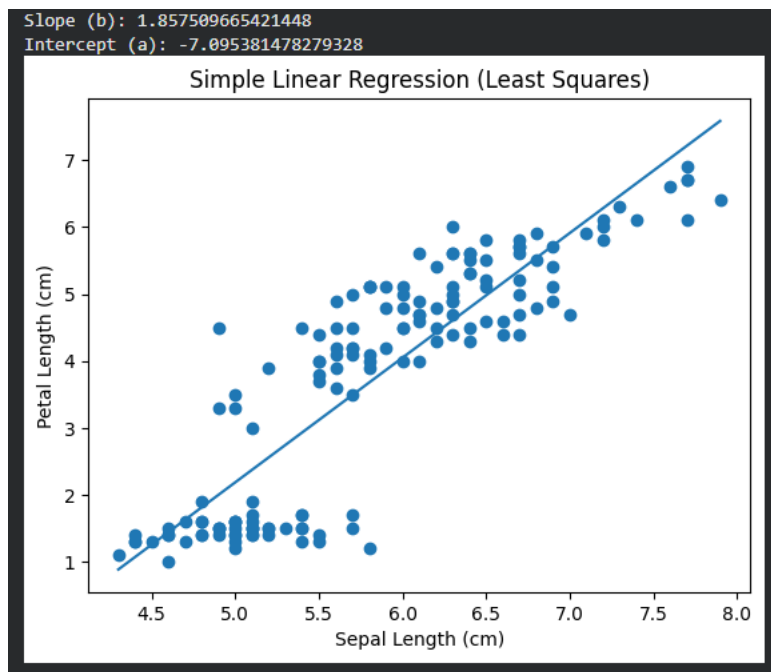
```
plt.title("Simple Linear Regression  
(Least Squares)")
```

```
plt.xlabel("Sepal Length (cm)")
```

```
plt.ylabel("Petal Length (cm)")
```

```
plt.show()
```

Output:



Result:

The Simple Linear Regression model was successfully implemented on the Iris dataset. The Least Square Method was used to calculate the best-fit regression line. The model was able to predict the dependent variable with reasonable accuracy, showing a linear relationship between the selected features. The performance was evaluated using metrics such as Mean Squared Error (MSE) and R^2 score.